1 DH description

The DH frames are assigned from the CuRo6R URDF, with frame $\{0\}$ at `base_link`, joint i rotating about z_{i-1}, and $q = 0$ matching the URDF home configuration. We use standard DH transforms:

$$^{i-1}T_i = R_z(\theta_i)\, T_z(d_i)\, T_x(a_i)\, R_x(\alpha_i). \tag{1}$$

Table 1 gives the resulting parameters and the geometric choices that determine them. The only fixed joint-angle offset is $\theta_3 - q_3 = \pi/2$.

Table 1: DH parameters of CuRo6R. Lengths are in metres; angular constants are in radians.

i	d_i	θ_i	α_i	a_i	Geometric origin of the parameters
1	1	q_1	$-\pi/3$	0.8	The two base heights add: $d_1 = 0.5 + 0.5$; the lateral offset is $a_1 = 0.8$. Joint 2's axis is its local y axis, so the 30° URDF roll gives $\alpha_1 = 30° - 90° = -60°$.
2	1.25	q_2	$+\pi/2$	2	The joint-3 offset resolves into 1.25 along z_1 and 2 along x_2. At home $x_2 = x_1$, and the orthogonal joint axes give $\alpha_2 = +90°$.
3	0.75	$q_3 + \pi/2$	$+\pi/2$	0	Moving 0.75 along z_2 reaches the intersection with joint 4's axis: O_3 in Fig. 1(a), hence $a_3 = 0$. Choosing $x_3 = y_2$ gives the +90° angle offset and $\alpha_3 = +90°$.
4	2.5	q_4	$-\pi/2$	0	The wrist centre lies $1.5 + 1 = 2.5$ along z_3 from O_3. Axes 4 and 5 intersect there; with $x_4 = x_3$ at home, $\alpha_4 = -90°$.
5	0	q_5	$+\pi/2$	0	Axes 5 and 6 meet at the wrist centre, so $O_5 = O_4$ and $d_5 = a_5 = 0$ [Fig. 1(b)]. Keeping $x_5 = x_4$ gives $\alpha_5 = +90°$.
6	1.5	q_6	0	0	The endpoint is 1.5 along z_5 from the wrist centre. Choosing $z_6 = z_5$ and $x_6 = x_5$ at home gives $a_6 = \alpha_6 = 0$.

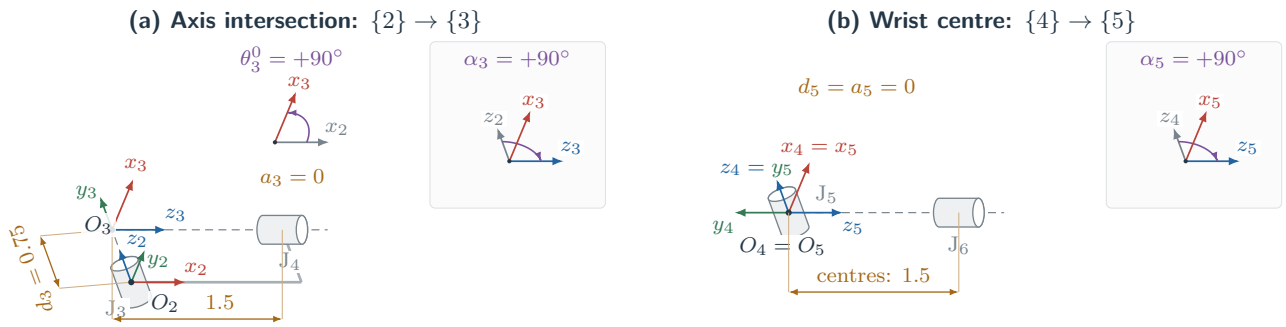


Figure 1: Frame placement at $q = 0$. (a) O_3 lies 1.5 m before the physical origin of joint 4 along its axis; this distance contributes to d_4. (b) $O_4 = O_5$ at the wrist centre; the 1.5 m between joints 5 and 6 contributes to d_6, while $d_5 = 0$. Cylinders mark physical joint origins and black dots mark DH origins; x, y, and z identify the frame axes.

Endpoint alignment. The endpoint E is `link_6`. Its origin coincides with O_6, while $x_E = z_6$, $y_E = x_6$ and $z_E = y_6$. The fixed alignment is therefore appended to the DH chain to recover the URDF endpoint pose:

$$^0T_E = \left(\prod_{i=1}^{6} {}^{i-1}T_i\right) {}^6T_E, \qquad {}^6T_E = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{2}$$

2 Analytical inverse kinematics

The axes of joints 4–6 intersect at the wrist centre, allowing position and orientation to be solved separately. We first determine the arm angles (q_1, q_2, q_3) from the desired wrist-centre position, then recover the wrist angles (q_4, q_5, q_6) from the desired orientation.

2.1 Wrist-centre positioning

Let R_d and p_d denote the desired orientation and position of `link_6` relative to `base_link`. Since the endpoint offset lies along the tool's x axis, the target wrist centre is

$$p_w = p_d - d_6 R_d e_x = (x_w, y_w, z_w)^T, \qquad e_x = (1, 0, 0)^T. \tag{3}$$

Throughout, $c_i = \cos q_i$ and $s_i = \sin q_i$.

1. Express the arm geometry in a shoulder-centred frame. For a candidate q_1, undo the base rotation, translate to the shoulder at $s = (a_1, 0, d_1)^T$, and undo the fixed tilt $\gamma = \pi/6$. The resulting auxiliary frame follows the base rotation and fixed tilt, but not the joint-2 rotation. With $h = z_w - d_1$ and $c_\gamma = \sqrt{3}/2$,

$$\begin{aligned} r(q_1) &:= R_x(-\gamma)\big(R_z(-q_1)p_w - s\big) = (X, Y, Z)^T, \\ X &= c_1 x_w + s_1 y_w - a_1, \\ Y &= c_\gamma(-s_1 x_w + c_1 y_w) + \tfrac{1}{2}h, \\ Z &= \tfrac{1}{2}(s_1 x_w - c_1 y_w) + c_\gamma h. \end{aligned} \tag{4}$$

These coordinates still depend on the unknown q_1.

In the moving URDF frame of `link_2`, the shoulder-to-elbow vector is $a = (a_2, d_2, 0)^T$. The elbow-to-wrist vector is $R_z(q_3)b_0$, where $b_0 = (d_4, 0, d_3)^T$ is expressed in the URDF frame of `link_3_d3`. Applying the shoulder rotation maps their sum into the auxiliary frame:

$$r(q_1) = R_y(q_2)\big(a + R_z(q_3)b_0\big) = R_y(q_2)\begin{pmatrix} a_2 + d_4 c_3 \\ d_2 + d_4 s_3 \\ d_3 \end{pmatrix}. \tag{5}$$

These frames differ from the DH frames: the geometric elbow rotation is q_3, whereas the DH parameter is $\theta_3 = q_3 + \pi/2$.

2. Eliminate the shoulder angle. Rotation about y preserves both the y component and the squared length. Equation (5) therefore gives two relations independent of q_2:

$$\begin{aligned} Y &= d_2 + d_4 s_3, \\ N := X^2 + Y^2 + Z^2 &= a_2^2 + d_2^2 + d_3^2 + d_4^2 + 2a_2 d_4 c_3 + 2d_2 d_4 s_3. \end{aligned} \tag{6}$$

Setting $K = d_2^2 - a_2^2 - d_3^2 - d_4^2$ yields

$$s_3 = \frac{Y - d_2}{d_4}, \qquad c_3 = \frac{N + K - 2d_2 Y}{2a_2 d_4}. \tag{7}$$

Here $a_2 = 2$ and $d_4 = 2.5$ are nonzero. These expressions determine s_3 and c_3 for each candidate q_1; enforcing $s_3^2 + c_3^2 = 1$ will determine which candidates are admissible.

3. Reduce the base-angle problem to a quartic. Since $R_x(-\gamma)$ preserves the norm, $N = \|r(q_1)\|^2 = \|R_z(-q_1)p_w - s\|^2$. Defining $N_0 = x_w^2 + y_w^2 + h^2 + a_1^2$ gives the compact expression

$$N = N_0 - 2a_1(x_w c_1 + y_w s_1). \tag{8}$$

Both N and Y are affine in (c_1, s_1). Substitution of (7) into $s_3^2 + c_3^2 = 1$ gives

$$4a_2^2(Y - d_2)^2 + (N + K - 2d_2 Y)^2 = 4a_2^2 d_4^2. \tag{9}$$

Introduce the half-angle variable $t = \tan(q_1/2)$, so that

$$D = 1 + t^2, \qquad c_1 = \frac{1 - t^2}{D}, \qquad s_1 = \frac{2t}{D}. \tag{10}$$

Writing $Y = \widetilde{Y}/D$ and $N = \widetilde{N}/D$, their quadratic numerators are

$$\begin{aligned} \widetilde{Y}(t) &= c_\gamma\big[y_w(1-t^2) - 2x_w t\big] + \tfrac{1}{2}hD, \\ \widetilde{N}(t) &= N_0 D - 2a_1\big[x_w(1-t^2) + 2y_w t\big]. \end{aligned} \tag{11}$$

Multiplying (9) by D^2 gives

$$\boxed{P(t) = 4a_2^2(\widetilde{Y} - d_2 D)^2 + (\widetilde{N} + KD - 2d_2\widetilde{Y})^2 - 4a_2^2 d_4^2 D^2 = 0.} \tag{12}$$

Each quantity being squared is quadratic in t, so P has degree at most four, generically four. Every real root supplies a candidate $q_1 = 2\arctan t$. The angle $q_1 = \pi$ must also be checked in the original constraint: it corresponds to $t = \infty$ and is absent from the finite half-angle parametrisation.

4. Recover and validate the arm configurations. For each candidate q_1, evaluate (4) and (7), then set $q_3 = \operatorname{atan2}(s_3, c_3)$ and $A = a_2 + d_4\cos q_3$. The remaining components of (5) are

$$X = c_2 A + s_2 d_3, \qquad Z = -s_2 A + c_2 d_3. \tag{13}$$

Thus

$$\begin{aligned} C_2 &= AX + d_3 Z = (A^2 + d_3^2)c_2, \\ S_2 &= d_3 X - AZ = (A^2 + d_3^2)s_2, \end{aligned} \qquad q_2 = \operatorname{atan2}(S_2, C_2). \tag{14}$$

The common factor is positive because $d_3 = 0.75$, so no division is needed before applying atan2.

The solution order is therefore $q_1 \to q_3 \to q_2$. All real-root candidates are retained for checking, rather than selecting one branch. Each recovered triple is verified through wrist-centre forward kinematics; angles are represented in $[-\pi, \pi]$, matching the URDF limits, and duplicates are removed modulo 2π. The remaining arm configurations are passed to the wrist-orientation problem. If leading polynomial coefficients vanish, the degree reduces; if $P \equiv 0$, q_1 is free and a representative of the continuous family can be chosen before back-substitution and validation.